F18CE – Midterm test 1

February 2022

Multiple choice questions

1. (1 point) Suppose that $\lim_{x\to a} g(x) = +\infty$. Then $\lim_{x\to a} \dfrac{f(x)}{g(x)} = 0$.

   ◯ True √ **False**

   Solution: Counterexample: $\lim_{x\to 0} \dfrac{1}{|\sin x|} = +\infty$, but $\lim_{x\to 0} \dfrac{|\sin x|}{|x|} = 1$. Basically, it depends on the function f.

2. (1 point) If all subsequences of a sequence converge, then the sequence itself converges.

   ◯ False √ **True**

   Solution: Take a sequence $\{a_n\} = \{a_1, a_2, a_3, a_4, \dots\}$ such that any subsequence converges. Then, $\{a_2, a_3, a_4 \dots\}$ is a subsequence of $\{a_n\}$ and it must converge. Therefore, $\{a_n\}$ converges too.

   In fact, you can prove that you only need a few well-chosen subsequences to converge, for the main sequence to converge. For instance, if $\{a_{2n}\}, \{a_{2n+1}\}, \{a_{3n}\}$ converge, then they converge to the same limit, and $\{a_n\}$ converges too (to the same limit). This is based on the fact that if a sequence converges, then all subsequences converge to the same limit.

3. (1 point) The function $f(x) = x \sin\left(\frac{1}{x^2}\right)$ is continuous on the interval $(-1, 1)$.

   √ **True** ◯ False

   Solution: The function $f(x) = x \sin(1/x)$ is not defined at $x = 0$, therefore it cannot be continuous. However, since the limit $\lim_{x\to 0} x \sin\left(\dfrac{1}{x}\right)$ exists, we can extend the definition of f to make it continuous:

   $$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \in (-1, 0) \cup (0, 1), \\ 0, & x = 0. \end{cases}$$

4. (1 point) $\lim_{x\to a} f(g(x)) = f(\lim_{x\to a} g(x))$.

   √ **True if f is continuous** ◯ True if g is continuous.

Solution: Take $g(x) = x$. To see that f needs to be continuous, take a function f that has a "jump" at the point $L = \lim x \to 0 g(x) = 0$, like the function

$$f(x) = \begin{cases} 1, & x \neq 0, \\ 2, & x = 0. \end{cases}$$

Then $\lim_{x\to 0} f(g(x)) = \lim_{x\to 0} f(x) = \lim_{x\to 0} 1 = 1$, but $f\left(\lim_{x\to 0} g(x)\right) = f\left(\lim_{x\to 0} x\right) = f(0) = 2$.

On the other hand, switch f and g to see that continuity of g is not necessary. If $f(x) = x$ and g is the jump function above, we have that $\lim_{x\to 0} f(g(x)) = \lim_{x\to 0} g(x) = 1$, and $f\left(\lim_{x\to 0} g(x)\right) = f(1) = 1$.

5. (1 point) Which of the following intervals will contain all of the function values of $f(x) = \frac{x^2-1}{3x-4}$ on the interval $[-4, 1]$?

◯ $\left[-\frac{15}{16}, 0\right]$ ◯ $[0, 1]$ √ $[-1, 1]$ ◯ $\left[-\frac{15}{16}, \frac{1}{4}\right]$

Solution: There are many ways to do this. Since it is a multiple choice question, you can simply check the value of the function at some points to exclude the intervals that don't work.

For instance, $f(1) = 0$ and $f(-4) = \frac{16-1}{-12-4} = -\frac{15}{16}$, so the interval must contain *at least* $[-15/16, 0]$. Then, $f(0) = \frac{-1}{-4} = \frac{1}{4}$, so the interval must contain $[-15/16, 1/4]$. Since we have two options left (the last two), we need to check if f is ever bigger than $1/4$. For instance, $f(1/2) = \frac{1/4-1}{3/2-4} = \frac{-3/4}{-5/2} = \frac{3}{10} > \frac{1}{4}$. Therefore, then only option is $[-1, 1]$.

6. (1 point) Let p be a polynomial. Then $\lim_{n\to\infty} \dfrac{p(n+1)}{p(n)} =$

◯ 0 √ 1 ◯ $\pm\infty$ ◯ Cannot be determined

Solution: Say that p is a polynomial of order m: $p(x) = a_0 + a_1 x + \cdots + x^m$. Then, $p(n) = a_0 + a_1 n + \cdots + n^m = n^m + (\text{stuff of order } < n^m)$, and $p(n+1) = a_0 + a_1 + a_1 n + a_2 + 2a_2 n + a_2 n^2 + \ldots n^m = n^m + (\text{stuff of order } < n^m)$. Then, $\lim_{n\to\infty} p(n+1)/p(n) = 1$. This is not a proof, but you were not asked to write one, you only had to guess the value of the limit.

7. (1 point) Which of the following sequences will *not* have any convergent subsequence?

◯ $a_n = n \sin\left(\frac{n\pi}{2}\right)$, ◯ $a_n = 1 + 2(-1)^n$, ◯ $a_n = 1 + (-1)^n 10^{(-1)^n n}$, √ $a_n = (-2)^n$.

Solution: The sequence given by $(-2)^n = (-1)^n 2^n$ is $\{-2, 4, -8, 16, -32, 64, -128, \ldots\}$ and it clearly doesn't have a convergent subsequence.

You can also observe that $\sin(n\pi/2) = \{1, 0, -1, 0, 1, 0, -1, 0, \ldots\}$ therefore a_{2n} is a convergent subsequence of $n\sin(n\pi/2)$. The second sequence is bounded, so it has a convergent subsequence (Bolzano–Weierstrass Thm), whereas for the third sequence we can consider the subsequence $\{a_{2n+1}\} = \{1 - 1/10^{2n+1}\}$, which converges to 0.

8. (1 point) If an upper bound exists, then it is unique.

◯ True √ **False**

Solution: The supremum and the infimum are unique, but not upper and lower bounds.

9. (1 point) Consider the sequence $\{a_n\}$ defined as $a_n = \frac{\sin(n)}{n}$. From the choices below, choose *all* correct statements that pertain to the sequence $\{a_n\}$.

   √ 2 **is an upper bound for the sequence.**

   ◯ If $n > 50$, then $|a_n| < 0.01$.

   √ **There is an integer** N **such that** $|a_n| < 0.0001$ **for all** $n > N$**.**

   ◯ There is an integer N such that $|a_n| > 0.0001$ for any $n > N$.

Solution: $\left|\frac{\sin n}{n}\right| \leq \frac{1}{n} \leq 1$, so 2 is an upper bound. In particular, $\lim_{n\to\infty} \frac{\sin n}{n} = 0$, therefore the third option is true by definition of limit (i.e. for $\varepsilon = 0.0001$). Regarding the second option, $a_{51} = \sin(51)/51 \sim 0.013 > 0.01$. The last option is false because it contraddicts the definition of limit.

10. (1 point) For $f(x) = \frac{1}{1+x^2}$, given $\varepsilon < 1$, what value of δ proves the continuity of f at $x = 1$? (You may assume $\delta < 1$.)

    ◯ $\delta = \varepsilon^2$, ◯ $\delta = \varepsilon$, √ $\delta = \frac{2}{3}\varepsilon$**.**

Solution:

$$\left|\frac{1}{1+x^2} - \frac{1}{2}\right| = \frac{|2-1-x^2|}{2(1+x^2)} = \frac{|1-x^2|}{2(1+x^2)} \leq |1-x|\frac{|1+x|}{2}.$$

If $\delta < 1$, then $|x-1| < \delta$ implies that $x + 1 < 2 + \delta < 3$, therefore

$$\left|\frac{1}{1+x^2} - \frac{1}{2}\right| \leq |1-x|\frac{|1+x|}{2} \leq \delta\frac{3}{2}.$$

Written problem

Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that

$$|f(x) - f(y)| \leq k|x-y|^{\frac{1}{2}}, \qquad \forall x, y \in \mathbb{R}, \tag{1}$$

for some $k > 0$. Write the ε-δ definition of continuous function on $\mathbb{R}$, and prove that f is continuous on $\mathbb{R}$.

Solution: A function $f : \mathbb{R} \to \mathbb{R}$ is *continuous at a point* a if for any $\varepsilon > 0$ there exists a $\delta = \delta(\varepsilon, a) > 0$ such that

$$|f(x) - f(a)| < \varepsilon,$$

for any $x \in \mathbb{R}$ such that $|x - a| < \delta$. A function is *continuous on* $\mathbb{R}$ if it is continuous at every point $a \in \mathbb{R}$.

Fix a point $a \in \mathbb{R}$ and $\varepsilon > 0$. Then, by (1), if $|x - a| < \delta$, we have that

$$|f(x) - f(a)| \leq k|x - a|^{\frac{1}{2}} < k\delta^{\frac{1}{2}}.$$

Therefore, we want to choose $\delta > 0$ such that $k\delta^{\frac{1}{2}} < \varepsilon$. This is equivalent to $k^2\delta < \varepsilon^2$, i.e. $0 < \delta < \dfrac{\varepsilon^2}{k^2}$. Hence, for any $x \in \mathbb{R}$ such that $|x - a| < \delta < \dfrac{\varepsilon^2}{k^2}$, we have that $|f(x) - f(a)| < k\delta^{\frac{1}{2}} < \varepsilon$, which proves continuity of f at any point $a \in \mathbb{R}$.

Observe that a function f that satisfies the property (1) is not only continuous, but also *uniformly continuous* on $\mathbb{R}$, as the $\delta = \delta(\varepsilon)$ does not depend on the choice of the point a at which we check continuity.